Validation and AI Use — Project 1, NVIDIA (NVDA)

Student: Drew Scheiderer · **Course:** FIN 43900, AI in Finance, Purdue, Fall 2026 **Status:** working source for `Validation-and-AI-Use.pdf`. Sections marked **[DREW]** are mine to complete. The Locked Changed-Input Record is filled by hand with AI closed.

> Coursework for a learning exercise. Not investment research and not financial advice.

Decision, version, and as-of boundary

Item	Value
Decision / user	Initiate, watch/defer, or do not initiate NVDA, for a buy-side investment committee holding no position
Horizon	3–5 year holding period; five-year explicit forecast FY2027E–FY2031E plus terminal value
Repository	https://github.com/dscheid7717/SPCX-research
Analysis commit	`9222448` (engine, app, frozen output). Final frozen commit: **[DREW: fill after last commit]**
Run timestamp	Frozen output generated 2026-10-15; cold-run log in `docs/cold-run-log.txt`
Source cutoffs	NVIDIA 10-K FY2026 (filed 2026-02-25) and 10-Q Q2 FY2027 (filed 2026-08-26); peer 10-Ks filed 2025-12-18 (AVGO) and 2026-02-04 (AMD); market data close 2026-10-14
Environment	Python 3.12.10; `requirements.txt` (numpy, pandas, matplotlib, scipy, yfinance, streamlit pinned ranges); finance engine is standard library only

Data and convention ledger

Material input	Source	As-of / availability	Definition / units	Transformation / exclusion	Risk
FY2026 revenue 215,938	10-K acc. 0001045810-26-000021, income statement	period 2026-01-25, filed 2026-02-25	USD M, GAAP	base year; hand-confirmed against Item 8 (Lab 10)	low
Opening balance sheet (assets 206,803, liabilities 49,510, equity 157,293)	10-K balance sheet	2026-01-25	USD M	mapped into 4 asset / 4 liability buckets; `check_opening_balance()` proves totals	low (checked)
H1 FY2027 revenue 177,837; Q2 96,221	10-Q acc. 0001045810-26-000075 (SEC XBRL `Revenues`)	period 2026-07-26, filed 2026-08-26	USD M	anchors FY2027E = H1 + 2 × Q2 = 370,279 (+71.47%)	medium: H2 is a judgment
H1 FY2027 gross margin 74.96%	10-Q (`GrossProfit` 133,299 / `Revenues`)	2026-07-26	ratio	forecast GM path starts at 75.0%	medium
FY2026 inventory write-down 4,000	10-K (XBRL `InventoryWriteDown`)	2026-01-25	USD M, in cost of revenue	normalisation: reported GM 71.07% → 72.92%	low
FY2026 equity-investment gains 8,918	10-K cash flow statement	2026-01-25	USD M, other income	excluded from forecast; stakes carried in bridge	medium
Cash + marketable securities 62,556	10-K balance sheet rows 16–17 (10,605 + 51,951)	2026-01-25	USD M	added in bridge; pro-forma holds treasury flat here	low (definition checked — see register)
Non-marketable equity securities 22,251	10-K balance sheet row 27	2026-01-25	USD M, carrying value	added in bridge (new in Edition B)	medium: carrying ≠ fair value
Debt 8,468	10-K Note 11 (999 + 7,469)	2026-01-25	USD M, carrying	subtracted in bridge; Jul-2026 debt of 33,366 disclosed, not used	medium: post-period issuance
Diluted shares 24,514	10-K EPS note	FY2026 weighted average	millions	per-share basis throughout	low (0.9% vs basic)
NVDA price $221.30	Yahoo Finance via yfinance	close 2026-10-14	USD	WACC equity weight, comps, decision	low
Risk-free 5.190%	^TNX via yfinance	close 2026-10-14	10-year UST yield	CAPM	low
Beta 2.216 (Blume 1.815)	own OLS, `nvidia/beta.py`, 5y monthly vs S&P 500	prices cached 2026-09-22	slope	CAPM; Blume in high case	**high: carries the answer**
ERP 5.0%	course convention	—	—	CAPM	high

Material input	Source	As-of / availability	Definition / units	Transformation / exclusion	Risk
Pre-tax cost of debt 2.92%	10-K Note 11, principal-weighted (Lab 06)	2026-01-25	rate	WACC (debt weight 0.16%)	negligible
Peer financials (AMD, AVGO)	10-K XBRL: AMD acc. 0000002488-26-000018; AVGO acc. 0001730168-25-000121	FY ends 2025-12-27, 2025-11-02	USD M	EV = price × diluted shares + debt − cash & ST investments; EBITDA = GAAP operating income + CF-statement D&A incl. acquisition amortisation	medium: AMD/AVGO D&A taken at tag level — **[DREW: hand-confirm against each cash flow statement]**
Peer prices AMD $570.82, AVGO $346.25	Yahoo Finance via yfinance	close 2026-10-14	USD	comps	low

Validation register

Claim / output	Failure mode	Test and expected result	Actual result	Disposition	Evidence
DCF engine is correct	arithmetic error in discounting / Gordon / bridge	**Known answer:** reproduce the course's 12 Lab 06 training answers ($27.50, nine grid cells, +1.78pp reverse DCF)	all 12 match to $0.005 / 0.01pp	pass	`test_project.KnownAnswerTests`
Pro-forma engine is correct	statement linkage error	**Known answer:** root Asbury training case prints published $291.75	matches	pass	`test_asbury_proforma_training_case`
Bridge cash is the right object	provider label ≠ filing line	**Independent definition check:** SEC XBRL `AvailableForSaleSecuritiesDebtSecurities` gives 39,520 for 2026-01-25; the 10-K balance sheet line "Marketable securities" gives 51,951	they differ by 12,431: the XBRL tag is the debt-securities note total, not the balance-sheet line	XBRL value **rejected**; filing line governs	10-K workbook `BALANCE_SHEET` row 17; this record
Value responds in the right direction	sign error in a driver	**Directional changed-input tests, written and committed before first execution (93a7ac9, 2026-10-15 12:39 ET):** beta ↑ → value ↓; FY2027E growth ↑ → value ↑; GM ↓ → value ↓; SBC as cash cost → value ↓; reset recovers base	all as predicted	pass	`DirectionalChangedInputTests`; `git show 93a7ac9`
Statements articulate	plugged or unrouted line	balance sheet balances every year; CFS ending cash = BS cash; base year = filed totals and revenue	max gap 0.00 on all three	pass	`ArticulationTests`; report "CHECKS"
DCF uses the pro-forma consistently	FCFF and FCFE built from different stories	FCFE = FCFF + after-tax net interest − debt repaid, every year	max gap 0.0000	pass	check "FCFF reconciles to the pro-forma's FCFE"
DCF boundaries	value moves the wrong way, or Gordon used where undefined	grid strictly ↓ in WACC and ↑ in g over 5 × 5; g ≥ WACC refuses	monotonic; refusal returns no value	pass	`DcfBoundaryTests`
Bridge reconciles	EV-to-equity arithmetic	EV + 62,556 + 22,251 − 8,468 = equity; × shares both ways	exact	pass	`BridgeTests`
Peer policy applied consistently	definition drift between companies	every company carries every field; fiscal ends within 120 days; positive EBITDA/EPS; EV/EBITDA recomputes; NVDA recomputed by hand	all pass; NVDA 40.3× by hand	pass	`PeerPolicyTests`
Highest-risk failure (model)	a broken sheet is valued silently	inject SBC-without-equity-credit and an opening mis-map; model must refuse, gap must equal SBC exactly; next run must be clean	refuses; gap = SBC; no leak	pass	`HighestRiskFailureTests`; app fault switch
Highest-risk failure (evidence)	a stale driver carried forward	compare Lab 10 FY2027E growth (30%) to the 10-Q filed 2026-08-26	H1 already +95.8% YoY; 30% would need H2 revenue to fall to 102,882	**Lab 10 driver failed**; replaced (Edition B)	`analysis.py` DRIVER_RECORD

Claim / output	Failure mode	Test and expected result	Actual result	Disposition	Evidence
Decision rule can be moved by evidence	rule that no downside can alter	first-run review: does any downside case change the action?	**failed** under the median-only rule (`93a7ac9`); after adding the earnings-persistence leg the low case and hyperscaler-pause what-if give DO NOT INITIATE	revised, disclosed	`9222448` commit message; `analysis.py` decision-rule block
Frozen output is reproducible	hand-edited or stale visible output	fresh `run_analysis()` equals `visible_output.json`	equal	pass	`ReproducibilityTests`
Clean environment reproduces	hidden local state	**cold run:** fresh clone, fresh venv, `pip install -r requirements.txt`, tests, analysis, diff against frozen report, app smoke test	run 1 **failed** at the app step (`streamlit.proto` missing: Windows long paths disabled, 176-character folder prefix); run 2 from a short path clean — 27/27 tests, report identical to frozen, app WATCH/DEFER $78.83, 291 s	failure diagnosed as environment, not code; README now tells Windows users to clone to a short path	`docs/cold-run-log.txt`

Locked Changed-Input Record — human prediction first; AI closed during execution

[DREW — to be completed by hand, AI closed, BEFORE the run.] Commit this file with only the precommit table filled, note the commit hash, then run the change and fill the second table. Suggested command (any one input):

```
cd nvidia
python -c "import analysis as a; i=a.base_inputs(); i['beta']=1.815; r=a.run_analysis(i); print(round(r['base']['valuation']['value_per_share'],2), r['decision'])"
```

Precommit before the run and before AI assistance on the prediction

Precommit timestamp	Repository commit	Material input	Old → new value/units	Expected output direction	Expected decision effect

Preserved execution result

Before/after output	Actual decision effect	Prediction reconciliation	Failure diagnosis or why action did not change	Evidence path
base $78.83 →				

Supporting history: the Lab 11 locked prediction (`dbf76fd`, 2026-09-29 15:34 ET) is a prior human-authored record on the Lab 10 model; it lacks the decision-effect fields this template requires, which is why a Project 1 record is made above.

Named AI-use register

Tool / model and version	Date	Material task	Output / claim used or considered	Independent check	Disposition	Effect on decision
ChatGPT / Codex; Claude Code	2026-09-08 → 2026-10-01	Lab 06–12 drafting and code, as disclosed in each lab write-up	Lab models and write-ups	known-answer reproduction each lab; hand checks recorded in the labs	accepted with the checks recorded there	built the Lab 06–11 evidence base
Claude Code (Claude Opus 5.5)	2026-10-15	Pull NVDA and peer filing facts from SEC XBRL company-facts	candidate value 39,520 for "marketable securities" (`AvailableForSaleSecuritiesDebtSecurities`)	10-K balance sheet, workbook `BALANCE_SHEET` row 17: 51,951	**rejected** — wrong object (note total, not the balance-sheet line)	none: bridge kept at 62,556

Tool / model and version	Date	Material task	Output / claim used or considered	Independent check	Disposition	Effect on decision
Claude Code (Claude Opus 5.5)	2026-10-15	Review of Lab 10 drivers against the Q2 FY2027 10-Q	claim that FY2027E growth of 30% is contradicted by H1 +95.8%	**[DREW: open the 10-Q income statement and confirm 177,837 / 90,805]**	**[DREW: accept / modify]**	base value $58 → $79
Claude Code (Claude Opus 5.5)	2026-10-15	Proposed Edition B driver values, low/high cases, what-ifs	values in `analysis.py` BASE_INPUTS, CASES, WHAT_IFS	each anchored to a cited 10-Q / 10-K figure in DRIVER_RECORD	**[DREW: accept / modify each — these are judgments I must own]**	sets the $49–$115 range
Claude Code (Claude Opus 5.5)	2026-10-15	Decision rule	first version could not be moved by any downside case	first run of the engine	**modified**: earnings-persistence leg added, disclosed in commit 9222448	low case now flips to DO NOT INITIATE
Claude Code (Claude Opus 5.5)	2026-10-15	Wrote `analysis.py`, `app.py`, `test_project.py`	engine, app and 27 tests	known answers, articulation checks, cold run; **[DREW: read every function before Video 2]**	**[DREW]**	delivery
Claude Code subagent	2026-10-15	Read the public course repo for Project 1 requirements	"the analytic rubric file is not in the public repo"	**[DREW: check Brightspace]**	qualified: true of the public repo only	none

Limitations, monitoring, and kill/escalation rules

Limitations

- The forecast opens on the FY2026 balance sheet (2026-01-25) and discounts from that date; the price is 2026-10-14. Rolling the base value forward roughly 0.7 year at WACC would raise it about 11% before deducting H1 distributions — far from closing the gap to price.
- WACC rests almost entirely on beta. Raw 2.216 vs Blume 1.815 is a ~200bp swing ($79 → $93).
- The reverse DCF puts the disagreement in the discount rate: the price needs a beta of 0.51 (cost of equity 7.74%) on my operating forecast, or FY2028–31 growth 45pp above my path.
- Two peers only; trailing multiples lag NVIDIA's growth; AMD's multiple is inflated by thin margins and acquisition amortisation.
- Scenario endpoints are judgments, not probabilities.

Monitoring and kill/escalation rules | Trigger | Metric and threshold | Window | Action | |---|---|---|---| | Initiate | NVDA close ≤ base-case value | daily | escalate to committee for initiation | | Downside evidence | any 10-Q with YoY revenue decline, or GM < 70% | each 10-Q (next: Q3 FY2027) | rerun; rule gives DO NOT INITIATE in the low-case path | | Relative value lost | NVDA trailing EV/EBITDA ≥ lowest admitted peer | each 10-K / price review | DO NOT INITIATE | | Model integrity | any check fails | every run | kill: no value is reported until fixed |